

Summary of Integration Test Scripts for the Rental Management System

Integration testing is essential for verifying those different components of an application work together seamlessly. In the Rental Management System, the following integration tests ensure that critical functionalities for tenants and landlords operate as intended.

1. Test: User Registration and Login

Purpose:

To verify that new users can successfully register and log in to the Rental Management System, ensuring that the user authentication process is functioning correctly.

Intended Verification:

This test confirms that the system correctly handles user registration and login processes. It ensures that user data is properly stored and that authenticated sessions allow access to user-specific content, verifying the integration between the registration, authentication, and dashboard components.

2. Test: Landlord Adds a New Property

Purpose:

To ensure that landlords can add new properties to the system and that these properties are correctly stored and retrievable from the database.

Intended Verification:

This test validates the integration of the property management system with user authentication. It ensures that landlords can add properties and that the system correctly handles and stores property information, including file uploads, demonstrating seamless interaction between these components.

3. Test: Tenant Views Properties and Sends Message

Purpose:

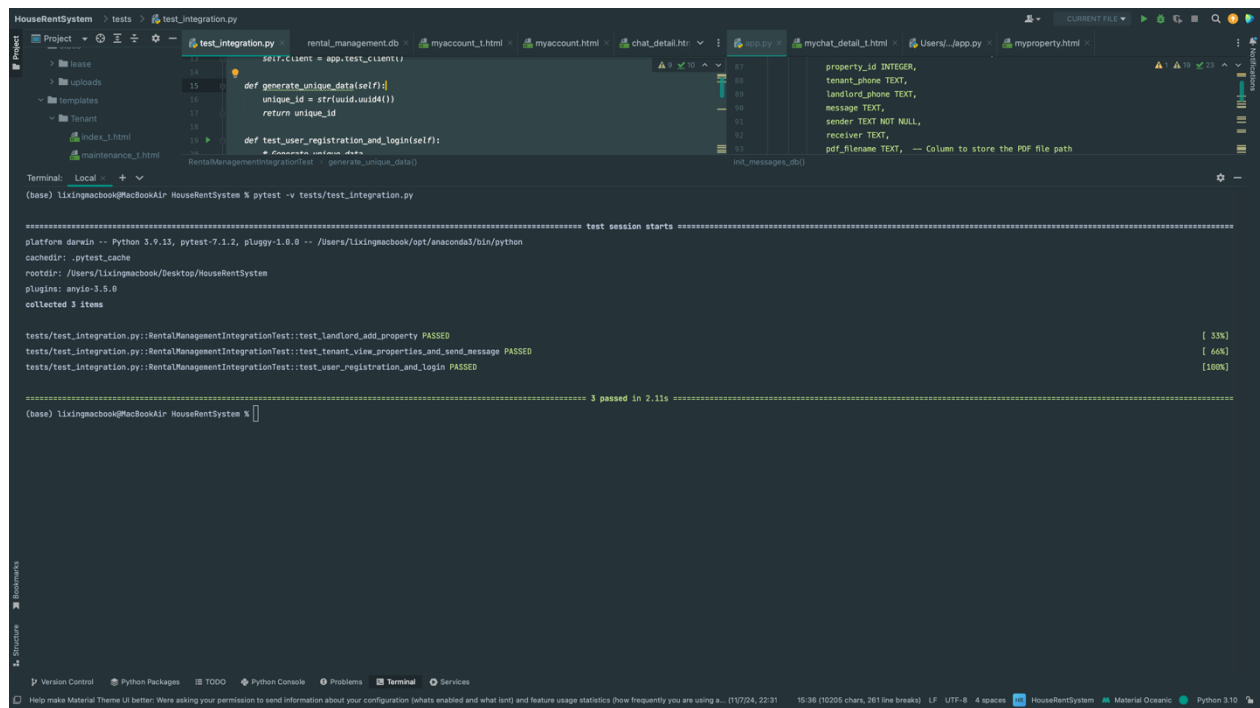
To verify that tenants can view available properties, access property details, and communicate with landlords through the messaging system.

Intended Verification:

This test ensures the proper integration of the property browsing, property details, and messaging components. It confirms that tenants can interact with properties and landlords as expected, and that the system accurately records and handles these interactions.

Overall Intention:

These integration tests are designed to simulate realistic user interactions within the Rental Management System, covering critical functionalities for both tenants and landlords. By testing the registration and authentication processes, property management, and communication features, the tests verify that the system's components interact correctly and reliably. This comprehensive testing approach helps identify and resolve potential issues, contributing to a robust and user-friendly application.



Step-by-Step Instructions

1. Navigate to Your Project Directory

Open your terminal or command prompt and navigate to the root directory of your Rental Management System project. Make sure you are in the root directory of the project (where requirements.txt and app.py are located) before running the following commands.

```
cd path/to/HouseRentSystem
```

2. Set Up a Virtual Environment (Optional but Recommended)

Creating a virtual environment helps manage dependencies and keeps your project environment isolated.

For Windows:

```
python -m venv venv
```

```
venv\Scripts\activate
```

For macOS/Linux:

```
python3 -m venv venv
```

```
source venv/bin/activate
```

3. Install Required Python Packages

Install all the required packages using pip. The main packages you need are:

- Flask
- Werkzeug
- uuid (standard library, no installation needed)
- pytest

You can install them using:

```
pip install Flask Werkzeug pytest
```

4. Ensure the Application Is Not Running

Before running the tests, make sure that the application server is not running. If it is, stop it to prevent database access conflicts.

5. Run the Integration Tests

You have two options to run the tests: using unittest directly or using pytest.

Using pytest

If you have pytest installed, you can run the tests with:

```
pytest tests/test_integration.py
```

6. Observe Test Output

The tests will run, and you will see output indicating which tests passed or failed.

- **Successful Test Output Example:**

```
=== test session starts ===
```

```
collected 3 items
```

```
tests/test_integration.py ...
```

```
=== 3 passed in X.XXs ===
```

- **Failed Test Output Example:**

If any tests fail, the output will provide details about the failures.